Lesson notes · for you

Reading data safely

Data science · Level 1 · Lesson 4

Time	Materials
30 min	A notebook and a pencil; A phone or tablet with Darsi, for the lab

Learning goals

- Look inside groups before comparing averages.
- Ask what else could explain a pattern.
- Share results without exposing a child.

Activity: Claim checkers

1. Warm-up (5 min): Read the claim “9 out of 10 pupils love our juice” and say what you would ask first.
2. Check (10 min): For each claim card, ask: how many people, what else could explain it, what data would check it?
3. Protect (10 min): Decide which class data is safe to share without names and which should stay private.
4. Wrap-up (5 min): Say your finding in one sentence with a number in it, then fill in the exit ticket.

Discussion

- Which claim was weakest?
- Why are small groups risky?
- What should never go on a class chart?

Check yourself

- Write one pattern from school life and one other thing that could explain it.

Try it this week

Find one claim with a number in an advert or online post. Ask what else could explain it and what data would check it, before you believe it or share it.

Worksheet

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Name _____ Class _____ Date _____

1 **What is the safest way to show parents how the class did?**

◯ The class average and spread ◯ A ranked list from best to worst

◯ Every pupil's mark, by name

2 **"9 out of 10 children love our juice!" What would you ask? Choose all that apply.**

☐ What colour is the bottle? ☐ How many children were asked? ☐ Who were they?

☐ What else could explain it?

3 **"9 out of 10 children love our juice!" The company asked 10 children. What percentage said yes?**

◯ 90% ◯ 99% ◯ 10% ◯ 9%

4 **It is fine to share a chart of everyone's test marks with their names.**

◯ True ◯ False

5 **Only one pupil in your class walks from the village. Why is sharing "village pupils' marks" risky?**

◯ Everyone would know whose marks they are ◯ Villages have no data

◯ The chart would be too big

✎ **What is one thing about you that you would not want shared in a school chart? Why?**

_____

_____

_____

Check your answers

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1 **The class average and spread**

The class picture tells parents enough without pointing at any child.

2 **How many children were asked?; Who were they?; What else could explain it?**

How many, who, and what else are the detective's questions.

3 **90%**

9 out of 10 is the same as 90 out of 100: 90%.

4 **False**

Marks are private. Share data without names, and take care with small groups.

5 **Everyone would know whose marks they are**

In a group of one, there is no hiding: the number belongs to one person.